University of Trier
Software Engineering Group
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Information Visualization
Summer Semester 2026
Exercise Sheet 5

Group work: Please keep in mind that the exercises are meant to prepare you for the exam.
ChatBots: Use your own brain as much and as often as possible.

Exercise 1 (Implementation of Graph Layout)

Implement force-directed graph layout in Java or Python based on the description of the spring-embedder method of Peter Eades as described on page 72ff in [1] and available in StudIP in file exercise05_spring_embedder.pdf. As a starting point for your implementation you should use the code provided in `exercise05_code.zip` in StudIP. It contains a data structure for graphs and their nodes, as well as the code to perform to control the steps of the simulation and draw the graph based on the current positions of its nodes. Your task is to implement the simulation steps by completing two functions which have empty function bodies in the provided source code.

In Python:

- `compute_forces_for_each_node(graph)`
- `compute_new_position_for_each_node(graph, delta)`

In Java:

- `void computeForces(Graph graph)`
- `void updatePositions(Graph graph, double delta)`

Reference:
[1]Kaufmann, M., & Wagner, D. (Eds.). (2003). Drawing graphs: methods and models. Springer.

Exercise 2 (Paper Critique)

Read the paper *"A Comparison of the Readability of Graphs Using Node-Link and Matrix-Based Representations"* (PDF available in StudIP) and write a critique about that article (300 to 400 words). Such a critique typically contains a brief summary but mainly focuses on your personal assessment. A critque should contain both positive and negative points. You should, however, always justify your statements. Tips for writing paper critiques can be found in the file exercise05_critiques.pdf in StudIP.